

ORAN ES SENIA, Algeria

WMO: 604900

Lat: 35.624N

Lon: 0.621W

Elev: 90

StdP: 100.25

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	3.9	-0.9	3.5	10.7	0.9	4.1	9.8	13.6	14.9	12.3	15.1	0.8	180	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	34.4	21.2	32.8	21.4	31.1	21.6	25.1	29.8	24.4	28.8	23.8	28.0	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.8	27.7	23.0	18.0	27.1	22.2	17.1	26.5	76.8	30.0	74.1	28.8	71.6	28.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.7	8.4	DB	0.1	39.2	1.2	2.0	-0.8	40.6	-1.5	41.8	-2.2	42.9	-3.1	44.4	(4)
			WB	-0.5	26.4	1.4	1.1	-1.6	27.1	-2.4	27.8	-3.2	28.4	-4.3	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	11.5	12.3	14.5	16.5	19.6	23.2	26.1	26.6	23.8	20.5	15.4	12.6
	DBStd	5.74	2.29	2.55	2.37	2.21	2.27	2.15	1.92	1.85	2.16	2.57	2.87	2.37
	HDD10.0	26	11	8	1	0	0	0	0	0	0	0	1	5
	HDD18.3	861	212	169	121	62	13	0	0	0	0	9	94	179
	CDD10.0	3159	57	72	140	196	298	396	498	516	414	324	164	84
	CDD18.3	953	0	1	2	8	52	146	239	257	164	76	8	1
	CDH23.3	7902	1	4	27	73	294	1065	2222	2536	1196	431	51	3
CDH26.7	2736	0	2	4	10	56	306	876	1031	347	96	8	1	
Wind	WSAvg	3.4	3.0	3.4	3.6	4.0	4.0	4.0	3.8	3.6	3.4	2.8	2.9	2.7
Precipitation	PrecAvg	366	46	44	41	31	26	6	1	2	15	33	51	56
	PrecMax	631	145	140	126	114	148	36	29	20	70	171	183	190
	PrecMin	172	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	101	33	31	33	24	29	8	4	4	15	34	39	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	23.0	26.1	28.0	30.9	34.5	37.3	37.2	33.9	32.2	27.8	22.9
		MCWB	13.8	14.1	15.9	16.8	17.9	20.4	20.7	21.3	20.9	19.1	16.2	14.9
	2%	DB	19.4	20.8	23.2	25.2	28.1	31.8	34.1	34.8	31.2	29.0	24.6	20.2
		MCWB	13.4	13.6	14.9	15.8	17.6	20.2	21.1	21.7	21.1	19.3	16.6	14.3
	5%	DB	18.0	19.1	21.3	23.3	26.2	29.8	32.2	32.8	29.9	27.1	22.6	18.9
		MCWB	12.9	13.2	14.4	15.5	17.6	20.0	21.5	22.1	21.2	19.4	16.1	13.8
	10%	DB	16.5	17.7	19.9	21.8	24.9	28.1	30.8	31.1	28.8	25.8	20.9	17.5
		MCWB	12.4	12.9	13.9	15.0	17.3	20.0	21.9	22.5	21.1	19.2	15.6	13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	15.8	17.6	18.5	20.7	23.7	25.7	26.4	24.4	22.4	19.5	16.5
		MCDB	18.7	19.8	23.5	24.5	26.7	29.0	30.5	30.8	29.0	26.8	23.3	20.6
	2%	WB	14.2	14.8	16.2	17.3	19.7	22.7	24.9	25.4	23.6	21.5	18.0	15.3
		MCDB	17.7	18.7	21.1	22.6	24.7	27.7	29.7	30.0	28.0	25.7	21.6	18.8
	5%	WB	13.6	14.0	15.2	16.6	19.0	22.0	24.2	24.7	23.0	20.7	17.2	14.5
		MCDB	16.9	17.7	19.6	21.3	23.7	26.7	28.6	29.2	27.2	24.9	20.9	17.9
	10%	WB	12.8	13.2	14.5	15.8	18.3	21.3	23.5	24.1	22.3	19.9	16.2	13.7
		MCDB	16.0	16.6	18.6	20.1	23.0	25.8	27.9	28.3	26.5	24.0	19.9	16.9
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.6	10.7	10.6	10.0	10.0	10.1	10.2	10.2	10.9	10.5	10.4
		MCDBR	12.4	13.0	13.7	13.7	12.6	12.6	12.9	13.0	11.8	12.8	12.7	12.0
		MCWBR	7.5	7.2	7.0	6.4	5.4	5.2	5.2	5.5	5.0	5.7	6.6	7.0
	5% WB	MCDBR	10.8	10.8	11.6	11.1	10.1	9.8	9.5	9.5	9.5	10.9	10.7	10.8
		MCWBR	7.1	6.6	6.4	5.9	4.9	4.7	4.6	4.6	4.8	5.5	6.1	6.9
Clear-Sky Solar Irradiance	taub	0.327	0.353	0.413	0.429	0.462	0.488	0.536	0.510	0.468	0.416	0.368	0.335	
	taud	2.490	2.413	2.217	2.198	2.129	2.060	1.933	2.008	2.137	2.277	2.400	2.484	
	Ebn at Noon	873	884	852	857	831	806	764	775	788	800	811	837	
	Edh at Noon	84	101	134	143	156	167	188	171	143	114	91	80	
All-Sky Solar Radiation	RadAvg	2.72	3.56	4.78	5.79	6.69	7.43	7.27	6.55	5.26	4.03	2.84	2.45	
	RadStd	0.23	0.34	0.32	0.42	0.46	0.26	0.18	0.21	0.19	0.27	0.23	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.21	N/A	N/A	N/A	+0.51	+0.48	N/A	N/A	+91	+64
(47)	Regional (10 neighbors)	+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page